Description

Image



Caption

1. Gold - Oregon Canyon Nugget (replica) © James St. John at flickr (CC BY-SA 2.0) 2. Gold ingots © AJC ajcann at flickr (CC BY-SA 2.0) 3. Gold Chains © Adam at flickr (CC BY-SA 2.0)

The material

Gold is the king of metals. Its name has entered the language as a symbol of perfection, achievement, good fortune and security: golden ages, golden ratios, golden rules, gold standards, golden crowns, golden girl, go for gold, gilded lives. When times are bad and investments insecure, investors turn to gold. Its rich color, its malleability, its resistance to almost all corrosion and its sheer scarcity combine to make it the most desirable of metals.
Some 90% of the global production of gold gets squirreled away as bullion and jewelry. The remaining 10% plays vital roles in the electronics industry as interconnects and as surface layers on connectors. Its high electrical conductivity and resistance to attack allow extreme miniaturization. Its use in dentistry, once large, is diminishing.

Compositional summary

>99.5%Au

General properties

Density		1.93e4	-	1.94e4	kg/m^3
Price	*	4.26e4	-	5.31e4	GBP/kg
Date first used		-6000			

Mechanical properties

Young's modulus		76	-	81	GPa
Shear modulus		26	-	30	GPa
Bulk modulus		148	-	180	GPa
Poisson's ratio		0.415	-	0.425	
Yield strength (elastic limit)		165	-	205	MPa
Tensile strength		180	-	220	MPa

Compressive strength	*	165	-	205	MPa
Elongation		2	-	6	% strain
Hardness - Vickers		50	-	70	HV
Fatigue strength at 10^7 cycles	*	70	-	110	MPa
Fracture toughness	*	60.1	-	66.5	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	0.0016	-	0.0021	

Thermal properties

Melting point		1.06e3	-	1.07e3	°C
Maximum service temperature	*	130	-	220	°C
Minimum service temperature		-273			°C
Thermal conductor or insulator?		Good conductor			
Thermal conductivity		305	-	319	W/m.°C
Specific heat capacity		125	-	135	J/kg.°C
Thermal expansion coefficient		13.5	-	14.5	µstrain/°C

Electrical properties

Electrical conductor or insulator?		Good conductor			
Electrical resistivity		2	-	3	µohm.cm

Optical properties

Transparency	Opaque

Critical Materials Risk

High critical material risk?	No

Durability: water and aqueous solutions

Water (fresh)	Excellent
Water (salt)	Excellent
Soils, acidic (peat)	Excellent
Soils, alkaline (clay)	Excellent
Wine	Excellent

Durability: acids

Acetic acid (10%)	Excellent
Acetic acid (glacial)	Excellent
Citric acid (10%)	Excellent
Hydrochloric acid (10%)	Excellent
Hydrochloric acid (36%)	Excellent
Hydrofluoric acid (40%)	Excellent

Values marked * are estimates.
ANSYS, Inc. provides no warranty for this data.

Nitric acid (10%)	Excellent
Nitric acid (70%)	Excellent
Phosphoric acid (10%)	Excellent
Phosphoric acid (85%)	Excellent
Sulfuric acid (10%)	Excellent
Sulfuric acid (70%)	Excellent

Durability: alkalis

Sodium hydroxide (10%)	Excellent
Sodium hydroxide (60%)	Excellent

Durability: fuels, oils and solvents

Amyl acetate	Excellent
Benzene	Excellent
Carbon tetrachloride	Excellent
Chloroform	Excellent
Crude oil	Excellent
Diesel oil	Excellent
Lubricating oil	Excellent
Paraffin oil (kerosene)	Excellent
Petrol (gasoline)	Excellent
Silicone fluids	Excellent
Toluene	Excellent
Turpentine	Excellent
Vegetable oils (general)	Excellent
White spirit	Excellent

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Excellent
Acetone	Excellent
Ethyl alcohol (ethanol)	Excellent
Ethylene glycol	Excellent
Formaldehyde (40%)	Excellent
Glycerol	Excellent
Methyl alcohol (methanol)	Excellent

Durability: halogens and gases

Chlorine gas (dry)	Excellent
Fluorine (gas)	Excellent

O2 (oxygen gas)	Excellent
Sulfur dioxide (gas)	Excellent

Durability: built environments

Industrial atmosphere	Excellent
Rural atmosphere	Excellent
Marine atmosphere	Excellent
UV radiation (sunlight)	Excellent

Durability: flammability

Flammability	Non-flammable

Durability: thermal environments

Tolerance to cryogenic temperatures	Excellent
Tolerance up to 150 C (302 F)	Excellent
Tolerance up to 250 C (482 F)	Excellent
Tolerance up to 450 C (842 F)	Excellent
Tolerance up to 850 C (1562 F)	Unacceptable
Tolerance above 850 C (1562 F)	Unacceptable

Geo-economic data for principal component

Annual world production, principal component	3e3			tonne/yr
Reserves, principal component	4.82e4			tonne

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	* 5.95e4	-	6.56e4	kg/kg
Embodied energy (virgin grade production)	* 4.1e5	-	4.52e5	MJ/kg
Water usage	* 2.61e5	-	2.89e5	l/kg

Material processing: energy

Metal casting, embodied energy	* 33.1	-	37	MJ/kg
Metal extrusion, hot, embodied energy	* 4.2	-	5.5	MJ/kg
Metal rolling and forging, embodied energy	* 2.55	-	3.25	MJ/kg
Wire drawing, embodied energy	* 13.1	-	14.5	MJ/kg
Metal powder forming, embodied energy	* 15.5	-	18.4	MJ/kg
Chemical vapor deposition (CVD), embodied energy	* 2.22e3	-	2.87e3	MJ/kg
Machining, coarse, embodied energy	* 15.1	-	16.8	MJ/kg
Machining, fine, embodied energy	* 16.1	-	18.1	MJ/kg
Machining, grinding, embodied energy	* 17.1	-	19.5	MJ/kg
Machining, laser, CO2, embodied energy	* 0.265	-	0.341	MJ/m

Material processing: climate change

Metal casting, climate change (CO2-eq)	* 2.23	-	2.5	kg/kg
Metal extrusion, hot, climate change (CO2-eq)	* 0.275	-	0.361	kg/kg
Metal rolling and forging, climate change (CO2-eq)	* 0.167	-	0.213	kg/kg
Wire drawing, climate change (CO2-eq)	* 0.86	-	0.95	kg/kg
Metal powder forming, climate change (CO2-eq)	* 1.23	-	1.46	kg/kg
Chemical vapor deposition (CVD), climate change (CO2-eq)	* 183	-	236	kg/kg
Machining, coarse, climate change (CO2-eq)	* 0.988	-	1.1	kg/kg
Machining, fine, climate change (CO2-eq)	* 1.05	-	1.19	kg/kg
Machining, grinding, climate change (CO2-eq)	* 1.12	-	1.28	kg/kg
Machining, laser, CO2, climate change (CO2-eq)	* 0.0218	-	0.0281	kg/m

Recycling and end of life

Recycle	✓			
Functional recycle	✓			
Climate change (CO2-eq), recycling	* 1.79e3	-	1.97e3	kg/kg
Embodied energy, recycling	* 8.2e3	-	9.04e3	MJ/kg
Recycle fraction in current supply	36			%
Combust for energy recovery	✗			
Landfill	✓			
Biodegrade	✗			
A renewable resource?	✗			

Supporting information

Design guidelines

At today's gold price, ore grades as low as 0.5 ppm are economic. The concentration of gold in some consumer electronics is greater than this, making old mobile phones and portable computers a viable resource for the metal. Gold prices are normally quoted in US$/Troy ounce. (1 kg = 32.151 Troy oz.)

Technical notes

Gold occurs in as native metal, usually in very small particles. The metal is isolated from sand or silt by "panning" - shaking in a water-filled sieve - using its high density to separate it. It is soluble in mercury from which it is recovered by distilling off the mercury. It dissolves in alkaline cyanide solutions from which it can be recovered by precipitation. All three methods are use in its extraction and recovery from used products.

Typical uses

jewelry, interconnects, printed circuit board edge connectors, electrical contacts, lining for chemical equipment, coinage, bullion, plating for space satellites, toning silver images in photography. Palladium, platinum, and silver may substitute for gold.

Links

ProcessUniverse

Producers

Reference

Values marked * are estimates.
ANSYS, Inc. provides no warranty for this data.